

FDTD: diffraction over objects

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1 Introduction

2 method

2.1 The equations

The equations to be solved are the following wave equation equations

$$\frac{\partial o}{\partial t} + \nabla p = 0 \quad (1)$$

$$\frac{\partial p}{\partial t} + c^2 \nabla \cdot o = 0 \quad (2)$$

or in cartesian coördinates

$$\frac{\partial o_x}{\partial t} + \frac{\partial p}{\partial x} = 0 \quad (3)$$

$$\frac{\partial o_y}{\partial t} + \frac{\partial p}{\partial y} = 0 \quad (4)$$

$$\frac{\partial p}{\partial t} + \frac{\partial o_x}{\partial x} + \frac{\partial o_y}{\partial y} = 0 \quad (5)$$

The fact that the simulated domain is limited means that there will be boundaries. This means there will always be reflections while the physical situation requires the waves to be able to leave the domain. To solve this problem an absorbing layer can be included so that by the time the wave reaches the edge of the domain there is no wave left to reflect seeing as it is already fully absorbed. Unfortunately this creates its own problems. The first one is that the sudden change of material from free space into an absorbing material will allow for both transmission and reflection. Thus even if there are no more reflections at the boundary there will still be reflections at the edge of the absorbing layer. The

second problem arises due to the angle of incidence. To solve the first one the absorbtion coëfficient can be gradually increased from zero in the simulation domain to desired maximum damping coëfficient in the material. This ramping up of the damping coëfficient in the material is given by

$$\kappa = \kappa_{max} \left(\frac{x}{d} \right)^5 \quad (6)$$

The second problem is solved through splitting the pressure in to an orthogonal and a parallel component. This doesn't neccecarly make sense for a scalar but this can simply be thought of as orthogonal and parallel contributions to the pressure caused by the orthogonal and parallel components of the velocity. This can be seen in eq ??

$$\frac{\partial o_x}{\partial t} + \frac{\partial p_x}{\partial x} = 0 \quad (7)$$

$$\frac{\partial o_y}{\partial t} + \frac{\partial p_y}{\partial y} = 0 \quad (8)$$

$$\frac{\partial p_x}{\partial t} + \frac{\partial o_x}{\partial x} = 0 \quad (9)$$

$$\frac{\partial p_y}{\partial t} + \frac{\partial o_y}{\partial y} = 0 \quad (10)$$

where the assumption has been made that the all 4 boudnrys are either parallel with with the y-axis or x-axis(i.e a recatngle).

The only thing still to be added to these equations is a damping coefficient. To get the desired result and full damping of the wave the damping coëfficients for the orthogonal components should be equal to eachother while the parallel components should be zero and this for each boundry. This was accomplished by solving the four equations with a damping coëfficient in the whole of the domain. Where the vertical damping coëfficients for the vertical boundrys is zero while the horizontal coëfficients are equal to eachother and ramp up from zero in the domain to a κ_{max} in the boundry layer. The exact oposite was done for the horizontal boundry's. This leads to the following equations to be solved:

$$\frac{\partial o_x}{\partial t} + \frac{\partial p_x}{\partial x} + \kappa_x o_x = 0 \quad (11)$$

$$\frac{\partial o_y}{\partial t} + \frac{\partial p_y}{\partial y} + \kappa_y o_y = 0 \quad (12)$$

$$\frac{\partial p_x}{\partial t} + \frac{\partial o_x}{\partial x} + \kappa_x p_x = 0 \quad (13)$$

$$\frac{\partial p_y}{\partial t} + \frac{\partial o_y}{\partial y} + \kappa_y p_y = 0 \quad (14)$$

with the follwoing damping coëfficients throughout space.

$$\kappa_x = 10 \left(\left(\frac{x - (a - d)}{d} \right)^5 H(x - (a - d)) - \left(\frac{x - d}{d} \right)^5 H(-x + d) \right) \quad (15)$$

$$\kappa_y = 10((\frac{y - (b - d)}{d})^5 H(y - (b - d)) - (\frac{y - d}{d})^5 H(-y + d)) \quad (16)$$

With a and b the width and height of the domain repsectivly and H the heaviside function. This can be seen in the heatmap in fig

2.2 The source

The equations described above will not produce any waves on their own, for this a source is neceary. In these simulation the domain of intrest lies for in $kd \in [0.1, 10]$ or thus $f \in [0.0028, 0.28]$. To this end the following source is used.

$$P_{source} = \sin(2\pi f_o) \exp(-\frac{(t - t_0)^2}{\sigma^2}) \quad (17)$$

This source can be seen in the time domain in fig. and in the frequency domain in fig. It is clear that the source forms a short band around f_0 spanning the values of kd .

2.3 finite difference equations

Now to solve these equations the pressure and velocity fields need to be discretised. This sampling can be done the same for all fields i.e evaluting all the fields in the same points(a collocated grid)

$$p_x = p_x|_{i,j}^l = p_x(idx, jdy, ldt) \quad p_y = p_y|_{i,j}^l = p_y(idx, jdy, ldt) \quad (18)$$

$$o_x = o_x|_{i+,j}^l = o_x(idx, jdy, ldt) \quad o_y = o_y|_{i,j}^l = o_y(idx, jdy, ldt) \quad (19)$$

but as will be further explained later, this is a bad idea because it leaves half the points untouched for pressure and the other half untouched for the velocitys. This is why a staggered grid was chosen so that the fields are evaluated as seen in

$$p_x = p_x|_{i,j}^l \quad p_y = p_y|_{i,j}^l \quad o_x = o_x|_{i+1/2,j}^{l+1/2} \quad o_y = o_y|_{i,j+1/2}^{l+1/2} \quad (20)$$

This grid can be seen in fig

Now to solve for these discretised functions the equations also need to be discretised. Seeing as the derivatives are based on continuous functions. This can be done through central finite difference based on the taylor expansion.

$$f(x + dx/2) = f(x) + f'(x)\frac{dx}{2} + \frac{1}{2}f''(x)\frac{dx^2}{4} + \dots \quad (21)$$

$$f(x - dx/2) = f(x) - f'(x)\frac{dx}{2} + \frac{1}{2}f''(x)\frac{dx^2}{4} + \dots \quad (22)$$

Substracting these two equations and rearranging gives

$$f'(x) = \frac{f(x + dx/2) - f(x - dx/2)}{dx} \quad (23)$$

Where it is clear that this aproximation is second order accurate seeing as only third order terms and higher are omitted. For derivatives in space this becomes

$$\frac{\partial p_x}{\partial x}|_{i+1/2,j}^l = \frac{p_x|_{i+1,j}^l - p_x|_{i,j}^l}{dx} \quad \frac{\partial p_y}{\partial y}|_{i,j+1/2}^l = \frac{p_y|_{i,j+1}^l - p_y|_{i,j}^l}{dy} \quad (24)$$

$$\frac{\partial o_x}{\partial x}|_{i,j}^{l+1/2} = \frac{o_x|_{i+1/2,j}^{l+1/2} - o_x|_{i-1/2,j}^{l+1/2}}{dx} \quad \frac{\partial o_y}{\partial y}|_{i,j}^{l+1/2} = \frac{o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j-1/2}^{l+1/2}}{dy} \quad (25)$$

and for derivatives in time this becomes

$$\frac{\partial p_x}{\partial t}|_{i,j}^{l+1/2} = \frac{p_x|_{i,j}^{l+1} - p_x|_{i,j}^l}{dt} \quad \frac{\partial p_y}{\partial t}|_{i,j}^{l+1/2} = \frac{p_y|_{i,j}^{l+1} - p_y|_{i,j}^l}{dt} \quad (26)$$

$$\frac{\partial o_x}{\partial t}|_{i+1/2,j}^l = \frac{o_x|_{i+1/2,j}^{l+1/2} - o_x|_{i+1/2,j}^{l-1/2}}{dt} \quad \frac{\partial o_y}{\partial t}|_{i,j+1/2}^l = \frac{o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j+1/2}^{l-1/2}}{dt} \quad (27)$$

When substituting this in the equation the following equations can be found

$$o_x|_{i+1/2,j}^{l+1/2} = o_x|_{i+1/2,j}^{l-1/2} - \frac{dt}{dx}(p_x|_{i+1,j}^l - p_x|_{i,j}^l) - \kappa_x o_x|_{i+1/2,j}^l \quad (28)$$

$$o_y|_{i,j+1/2}^{l+1/2} = o_y|_{i,j+1/2}^{l-1/2} - \frac{dt}{dx}(p_x|_{i,j+1}^l - p_x|_{i,j}^l) - \kappa_y o_y|_{i,j+1/2}^l \quad (29)$$

$$p_x|_{i,j}^{l+1} = p_x|_{i,j}^l - \frac{dt}{dx}(p_x|_{i+1/2,j}^{l+1/2} - o_x|_{i+1/2,j}^{l-1/2}) - \kappa_x p_x|_{i,j}^{l+1/2} \quad (30)$$

$$p_y|_{i,j}^{l+1} = p_y|_{i,j}^{l-1} - \frac{dt}{dx}(o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j+1/2}^{l-1/2}) - \kappa_y p_y|_{i,j}^{l+1/2} \quad (31)$$

Where each next time step can be calculated based on the previous step. There is still a problem however seeing as for the damping one needs to know the value of o_x and o_y at time steps l and p_x and p_y at time steps $(l+1/2)dt$ eventhough these values are not evaluated at those points in the grid. To solve this issue interpolation is used as can be seen in eq.

$$\kappa_x o_x|_{i+1/2,j}^l = \kappa_x \frac{o_x|_{i+1/2,j}^{l-1/2} + o_x|_{i+1/2,j}^{l+1/2}}{2} \quad (32)$$

$$\kappa_y o_y|_{i,j+1/2}^l = \kappa_y \frac{o_y|_{i,j+1/2}^{l-1/2} + o_y|_{i,j+1/2}^{l+1/2}}{2} \quad (33)$$

$$\kappa_x p_x|_{i+1/2,j}^l = \kappa_x \frac{p_x|_{i,j}^l + p_x|_{i,j}^{l+1}}{2} \quad (34)$$

$$\kappa_y p_y|_{i+1/2,j}^l = \kappa_y \frac{p_y|_{i,j}^l + p_y|_{i,j}^{l+1}}{2} \quad (35)$$

Now each next time step can be calculated based on the previous step. This means only an initial condition for the field in each point is required and the field through out time and space can be calculated. This initial value is chosen as zero in each point. However with these equations the field would remain zero seeing as a source is still to be added. To do this the value of the source at each time step is divided by two and added to both pressure equations(at the position of the source ofcourse). This way the total pressure at each time is still the pressure of the source and this total pressure is equally divided between the two contributions. After including the source and interpolation these are the final equations

$$(1 + \kappa_x \frac{dt}{2}) o_x|_{i+1/2,j}^{l+1/2} = (1 - \kappa_x \frac{dt}{2}) o_x|_{i+1/2,j}^{l-1/2} - \frac{dt}{dx} (p_x|_{i+1,j}^l - p_x|_{i,j}^l) \quad (36)$$

$$(1 + \kappa_y \frac{dt}{2}) o_y|_{i,j+1/2}^{l+1/2} = (1 - \kappa_y \frac{dt}{2}) o_y|_{i,j+1/2}^{l-1/2} - \frac{dt}{dx} (p_x|_{i,j+1}^l - p_x|_{i,j}^l) \quad (37)$$

$$(1 + \kappa_x \frac{dt}{2}) p_x|_{i,j}^{l+1} = (1 - \kappa_x \frac{dt}{2}) p_x|_{i,j}^l - \frac{dt}{dx} (p_x|_{i+1/2,j}^{l+1/2} - o_x|_{i+1/2,j}^{l-1/2}) \quad (38)$$

$$(1 + \kappa_y \frac{dt}{2}) p_y|_{i,j}^{l+1} = (1 - \kappa_y \frac{dt}{2}) p_y|_{i,j}^{l-1} - \frac{dt}{dx} (o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j+1/2}^{l-1/2}) \quad (39)$$

2.4 stability and accuracy

To make sure the solution is stable the time step is chosen in such a way that combined with the spatial steps the courant condition is fulfilled.

$$CN \leq 1 \quad (40)$$

with CN the courant number

$$CN = \sqrt{c^2 dt^2 (\frac{1}{dx^2} + \frac{1}{dy^2})} \quad (41)$$

On the other hand the dt must be as big as possible for accuracy and so dt is chosen such that

$$CN = 1 \quad (42)$$

2.5 finite difference boundary conditions

2.5.1 empty

With the equations described in the section above the simulation of an empty domain can now be calculated. However boundry conditions on the edge of the simulation domain still need to be implemented seeing as the domain is still finite. This boundary condition was chosen as perfectly hard. This is implemented by leaving all the values at the edges at zero.

2.5.2 Infinitely thin structure

For the infinitely thin structure the parameters of the empty simulation can be reused but this time a perfectly hard floor and wall need to be included. This can be done by assuming the plane representing the floor is at a plane where o_y is discretised and to put o_y on this plane equal to zero at each time step. To include the wall, the assumption is made that it lies in the plane where o_x is discretised. Then the values of o_x between the ground and the height of the wall in this plane are put equal to zero at each step. This can be seen in the following formulas

$$o_y|_{i,J+1/2}^{l+1/2} = 0 \quad (43)$$

$$o_x|_{I+1/2,j}^{l+1/2} = 0 \quad (44)$$

with $J+1/2$ the index of the ground plane, $I+1/2$ the index of the wall plane, j the indexes between J and the index at the top of the wall and i spanning all indexes in the domain.

2.5.3 Thick object

The procedure to include a hard wall and a hard thick object are completely analogous to the infinitely thin structure. The only difference is that now there are two walls. These are implemented in the exact same way as the one wall in the previous section. And lastly a "ceiling" needs to be implemented which is done in the same way as the floor but at a different height and instead of spanning the entire horizontal space, only the x values between the walls is considered.

$$o_y|_{i,J_{ground}+1/2}^{l+1/2} = 0 \quad (45)$$

$$o_x|_{I1+1/2,j}^{l+1/2} = 0 \quad (46)$$

$$o_x|_{I2+1/2,j}^{l+1/2} = 0 \quad (47)$$

$$o_y|_{ind,J_{ceiling}+1/2}^{l+1/2} = 0 \quad (48)$$

with $J_{ground}+1/2$ the index of the ground plane, $I1+1/2$ and $I2+1/2$ the index of the left wall- and right wall plane respectively, $J_{ceiling}+1/2$ the index of the ceiling plane, ind the indexes spanning from the left wall to the right wall and j the indexes going from the ground to the ceiling.

To implement a real impedance one needs to take a look at the update equations on the boundary for example on the left wall it looks like

$$o_x|_{I1+1/2,j}^{l+1/2} = o_x|_{I1+1/2,j}^{l-1/2} - \frac{2dt}{dx}(p|_{I1+1/2,j}^l - p|_{I1,j}^l) \quad (49)$$

Where a one sided finite difference was used seeing as the pressure beyond the boundary doesn't exist. Using the real impedance relation

$$p|_{I1+1/2,j}^l = Z o_x|_{I1+1/2,j}^l \quad (50)$$

the pressure on the boundry can be found in terms of the velocity on the boundry which is ideal seeing as it is a boundry where o_x is being discretised. However it is not ideal that the velocity needs to be evaluated at a time step ldt . This problem can be solved through interpolation as follows

$$o_x|_{I1+1/2,j}^l = \frac{o_x|_{I1+1/2,j}^{l+1/2} + o_x|_{I1+1/2,j}^{l-1/2}}{2} \quad (51)$$

after substituting this in and rearranging the following formula is achieved

$$(1 + Z \frac{dt}{dx}) o_x|_{I1+1/2,j}^{l+1/2} = (1 - Z \frac{dt}{dx}) o_x|_{I1+1/2,j}^{l-1/2} + \frac{2dt}{dx} p|_{I1,j}^l \quad (52)$$

and analougously for the other boundry's

$$(1 + Z \frac{dt}{dx}) o_x|_{I2+1/2,j}^{l+1/2} = (1 - Z \frac{dt}{dx}) o_x|_{I2+1/2,j}^{l-1/2} - \frac{2dt}{dx} p|_{I2,j}^l \quad (53)$$

$$(1 + Z \frac{dt}{dx}) o_y|_{ind,J_{ceiling}-1/2}^{l+1/2} = (1 - Z \frac{dt}{dx}) o_y|_{ind,J_{ceiling}-1/2}^{l-1/2} - \frac{2dt}{dx} p|_{J_{ceiling},j}^l \quad (54)$$

Where the oposite one sided difference is used in combination with the following impedance relationships

$$p = -Z o_x \quad (55)$$

$$p = -Z o_y \quad (56)$$

seeing as the direction of the impinging velocity is now against the unit axes.

2.5.4 Tilted boundry's

To approach tilted perfectly hard boundry condition a first attempt was made by trying to aproximate it with a "stair case" The boundry conditions in this case look like

$$o_y|_{index,4index-1/2+J_{ground}}^{l+1/2} = 0 \quad (57)$$

$$o_x|_{index+1/2,4index+J_{ground}}^{l+1/2} = 0 \quad (58)$$

$$o_x|_{index+1/2,4index+1+J_{ground}}^{l+1/2} = 0 \quad (59)$$

$$o_x|_{index+1/2,4index+2+J_{ground}}^{l+1/2} = 0 \quad (60)$$

$$o_x|_{index+1/2,4index+3+J_{ground}}^{l+1/2} = 0 \quad (61)$$

for the positive slope and

$$o_y|_{index+middle+I1+1,4middle+3-4index}^{l+1/2} = 0 \quad (62)$$

for the negative slope. Where $index$ denotes all the indices between zero and $middle$, the amount of points between the start of the obstacle and the middle. In this aproximation o_x represents "vertical parts" of each staircase step and o_y the horizontal part of every staircase step.

3 Infinitely Thin Structure

3.1 Analytical diffraction around wedge

The analytical solution for the infinitely thin wall can be calculated by using the following formula from the unified diffraction theory

(63)

This is however not the full contribution. Seeing as this is meant for a source and an obstacle in free space while our set up has a ground plane. This plane can be removed if another virtual source and wedge is added at the perfect reflected position (around the ground plane). This introduces 2 wedges and 2 sources and 4 contributions. When adding all these up one gets the analytical solution.

3.2 Results

When comparing the source with the observation it is clear that there is a time delay due to the field having to arrive at the observation point. And that there is decrease in amplitude seeing a lot of the power is radiated away in other directions. In the frequency domain...

4 Thick Object with Finite Impedance

4.1 Results

5 Tilted Boundaries

5.1 Finite Impedance

5.2 Refinement

In order to better model the non-cartesian surface, we included a grid refinement around the tilted boundaries.

This was achieved by making dx and dy functions of respectively x and y . As a first attempt both were made a step-function, in which around the boundaries they are both half of what they are elsewhere. This however results in reflections at the discontinuity in dx or dy , which can be seen when looking at an animation of a short Gaussian pulse. Next, to remove these reflections, a buffer zone was added in both x - and y -direction in which dx and dy are linearly interpolated (see fig ??).

Both these were in practice implemented by taking the original discretisation coordinates, and transforming these, either partwise linearly (for the

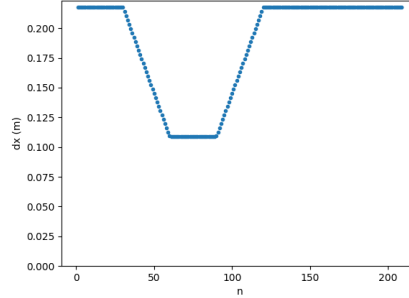


Figure 1: dx after refinement (linear)

step-function in dx or dy), or with a parabolic part in the bufferzone (for the linear dx and dy).

There were dx or dy is not constant however, this results in an error in the update scheme. As, for example, the discretised o_x may no longer be halfway between two p 's. This means that in the update equations of o_x , it is erroneous to use $\frac{dp}{dx}$ as calculated by differentiating two neighbouring p 's. So an interpolation was made between three values of $\frac{dp}{dx}$ at the halfway point between two p 's: two closest neighbours to the left, the two direct neighbours, and those to the right. Since there are three values, a parabolic interpolation was optimal. At the edges, where there are only two values to interpolate between, a linear interpolation was used instead. To update o_x , the interpolated $\frac{dp}{dx}$ at the location of o_x (after the coordinate transformation) was used.

Some results from the refinement and comparison with without: INSERT HERE.

5.3 Non Cartesian Grid

Another route that was explored is to tilt the grid. In this way the surface of the tilted boundary becomes flat on one side, thus removing the problems that arise from a staircase-like boundary. The left-side was chosen to be flat, as the most important reflections occur there.

Instead of working with x - and y -coordinates, a switch is made to u - and v -coordinates. These are related via equations ?? and ??, in which θ is the angle between the v -axis and the x -axis.

$$x = u + v \cos \theta \quad (64)$$

$$y = v \sin \theta \quad (65)$$

In figure ?? the basis-vectors are given. Note that $e_x = e_u$, but $x \neq u$. In this figure another vector, e_w , is introduced. This vector is perpendicular to e_v . With this figure the relations given in equations ?? and ?? are easily derived.

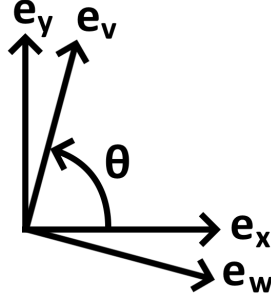


Figure 2: Important unit vectors

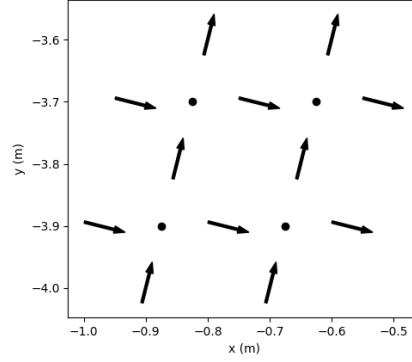


Figure 3: Discretised fields

$$\vec{e}_w = \sin \theta \vec{e}_x - \cos \theta \vec{e}_y \quad (66)$$

$$\vec{e}_v = \cos \theta \vec{e}_x + \sin \theta \vec{e}_y \quad (67)$$

There is still an important choice remaining, namely, which directions of $\vec{o}$ to store. A first idea might be to choose $o_u = \vec{e}_u \cdot \vec{o}$ and $o_v = \vec{e}_v \cdot \vec{o}$, as analogous to the cartesian grid. Remember however that the goal is to implement a tilted perfectly hard boundary, which has as boundary condition that $o_n = 0$, with n the normal of the boundary. For this boundary condition to be implemented simply, o_w and o_v are chosen, in which $\vec{e}_w$ is the normal of the tilted boundary (on the left side). An added benefit is that now both stored directions of $\vec{o}$ are orthogonal to eachother, so no data overlaps. Figure ?? shows the location of a few p , o_w and o_v 's.

5.3.1 Update Scheme

Now the update scheme can be adjusted for the tilted grid. To update p , the divergence of $\vec{o}$ is needed at the location of p , with $\nabla \cdot \vec{o} = \frac{do_w}{dw} + \frac{do_v}{dv}$. The following quantities are known:

$$\frac{do_v}{dv}|_{v_0+\Delta v/2} \cdot \Delta v \approx o_v|_{v_0+\Delta v} - o_v|_{v_0},$$

$$\frac{do_w}{du}|_{u_0+\Delta u/2} \cdot \Delta u \approx o_w|_{u_0+\Delta u} - o_w|_{u_0}.$$

$\frac{do_w}{dw}$ requires some more attention. From equations ?? and ??, the following can be derived:

$$\vec{e}_w = \frac{1}{\sin \theta} (\vec{e}_u - \cos \theta \vec{e}_v).$$

From which

$$\frac{do_w}{dw} = \vec{e}_w \cdot \nabla o_w = \frac{1}{\sin \theta} \left(\frac{do_w}{du} - \cos \theta \frac{do_w}{dv} \right)$$

follows. So the missing piece is $\frac{do_w}{dv}$, which can only be interpolated at p 's location by using

$$\frac{do_w}{dv}|_{v_0} \cdot 2\Delta v \approx o_w|_{v_0+\Delta v} - o_w|_{v_0-\Delta v},$$

combined with

$$o_w|_{u_0+\Delta u/2} \approx \frac{o_w|_{u_0} + o_w|_{u_0+\Delta u}}{2}.$$

For the update equations of $\vec{o}$ an analogous approach is followed.

In a hand-waving approach a rough approximation was made for the Courant Number. Starting from the standard form, and substituting $dx = du + dv \cos \theta$ and $dy = dv \sin \theta$ (derived from equations ?? and ??), equation ?? is calculated.

$$\text{CN} = c^2 dt^2 ((du + dv \cos \theta)^{-2} + (dv \sin \theta)^{-2}) \quad (68)$$

For stability to be achieved, $\text{CN} \approx 0.8$ is the maximum (for $\tan \theta = 4$). The adjusted update scheme involves a lot of interpolation of missing values, which decrease the stability.

5.3.2 Perfectly Matched Layers

$$\frac{dp_{\perp}}{dt} + c^2 \frac{do_{\alpha}}{d\alpha} + \kappa_{\perp} p_{\perp} = 0 \quad (69)$$

$$\frac{dp_{\parallel}}{dt} + c^2 \frac{do_{\gamma}}{d\gamma} + \kappa_{\parallel} p_{\parallel} = 0 \quad (70)$$

The perfectly matched layers in the tilted grid become a bit more complex. This especially for the upper PML. The update equations for $p = p_{\perp} + p_{\parallel}$

are given by equations ?? and ?. Looking at the upper PML, so that the perpendicular direction is $\vec{e}_y$ and the parallel one is $\vec{e}_x$, these equations become:

$$\frac{dp_{\perp}}{dt} + c^2 \frac{do_y}{dy} + \kappa_{\perp} p_{\perp} = 0,$$

$$\frac{dp_{\parallel}}{dt} + c^2 \frac{do_x}{dx} + \kappa_{\parallel} p_{\parallel} = 0.$$

By defining $\vec{p} = p_{\perp} \vec{e}_y + p_{\parallel} \vec{e}_x$ and $\vec{\kappa} = \kappa_{\perp} \vec{e}_y + \kappa_{\parallel} \vec{e}_x$, these can be rewritten in one equation as follows

$$\frac{d}{dt} \vec{e}_i \cdot \vec{p} + c^2 \vec{e}_i \cdot \nabla \vec{o} \cdot \vec{e}_i + |\vec{e}_i \cdot \vec{\kappa}| \vec{p} \cdot \vec{e}_i = 0$$

The absolute value of $\vec{e}_i \cdot \vec{\kappa}$ is important: if we chose $-\vec{e}_y$ instead of $\vec{e}_y$ as the perpendicular direction, the same equation should be gotten. Intuitively it is also non-sensical to have a negative damping-coefficient, as this would amplify. By now substituting $\vec{e}_i = \vec{e}_w$ or $\vec{e}_i = \vec{e}_v$, the equations for p_w and p_v are received.

It is important to note that this last step is not valid. Since this is an equation involving tensors, it cannot be generalized to any vector starting from two basis-vectors. This method, however, results in easy to implement PML's and, by visual validation, results in correct absorption.

5.3.3 Perfectly Hard Boundaries

The tilted boundary on the left side of the triangle is, as mentioned above, easy to implement. It just requires setting $o_w = 0$. The right-hand side isn't as simple to implement correctly (ie. such that only $o_n = 0$), instead a slightly incorrect alternative was used, setting all components of $\vec{o}$ to zero. Lastly the ground-plane had to be implemented. Neither of the simple approaches above are viable here, as $o_n = o_y$ is not stored directly so can't be set to zero, and neither can all components of $\vec{o}$ be set to zero. This last one is due to the fact that o_w and o_v aren't discretised at the same location, so that in reality the incoming wave will either see $o_w = 0$ or $o_v = 0$.

With this there is only one solution remaining, calculate o_y at the boundary by interpolation and remove this part from the stored o_v . In practice the opposite was done, o_x was calculated and o_v was set to only include this part. Approximating o_w at the ground by the one just above it, and using $o_x = o_w \sin \theta + o_v \cos \theta$, yields o_x . This boundary condition does not prevent waves from entering the boundary, which was mitigated by setting p equal to zero under the ground.

Finally another approach was tried. The goal is for o_y to be zero at the ground. Looking at its update equation,

$$\frac{do_y}{dt} + \frac{dp}{dy} = 0,$$

and using $o_y = 0$ for all t at $y = 0$, it follows that $\frac{dp}{dy} = 0$ at $y = 0$. As done before, this derivative can be written in terms of derivatives to u and v .

The derivative towards v is easily approximated at the ground by using central difference, but towards u requires interpolation. From this equation p just below the ground can be set such that the boundary condition is achieved. Sadly this method turned out to be very unstable and it was not useable.

Boundaries with finite impedance were not studied for the tilted grid.

6 Conclusion

... The tilted grid makes the implementation of the tilted boundary easier and more accurate. But comes with a few drawbacks. Mainly it becomes less stable, leading to smaller timesteps and more phase-error. Besides this it makes the implementation of the PML and ground more complex. TODO: summarize results

7 Conclusion